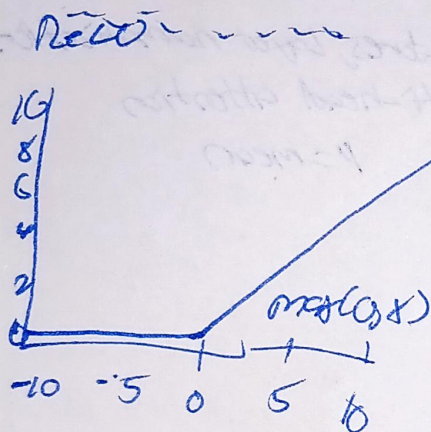


GELU Activation Function

o Implement a small NN sub-module that is a part of the LLM transformer block

o Two activation functions commonly implemented in LLMs

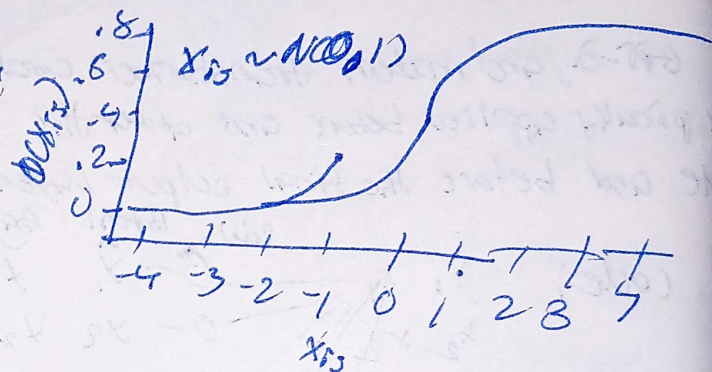
GBLU $\rightarrow X * \phi(x)$ $\rightarrow$ CDF of standard Gaussian distribution
 $\rightarrow$ cumulative distribution function
 SWIGLU standard Gaussian CDF



o Non-differentiable at $x=0$, and $x < 0$, its equal to 0

o Leads to dead neuron problems
 $\rightarrow$ if an output is -, and ReLU is applied, output is 0 and stays 0

o Don't contribute to learning process.

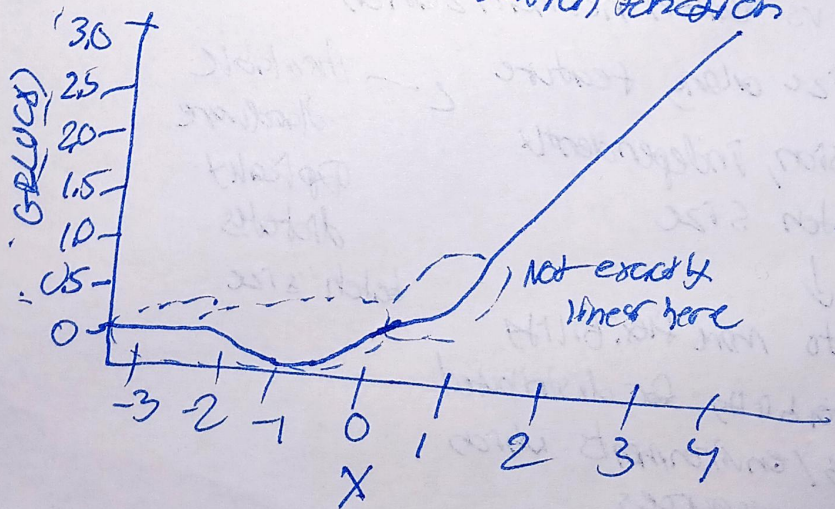


o Here is GELU Activation Function approximation used for training GPT-2

$$\text{GELU}(x) \approx 0.5 * x * (1 + \tanh(\sqrt{2/\pi} * (x + 0.044715 * x^3)))$$

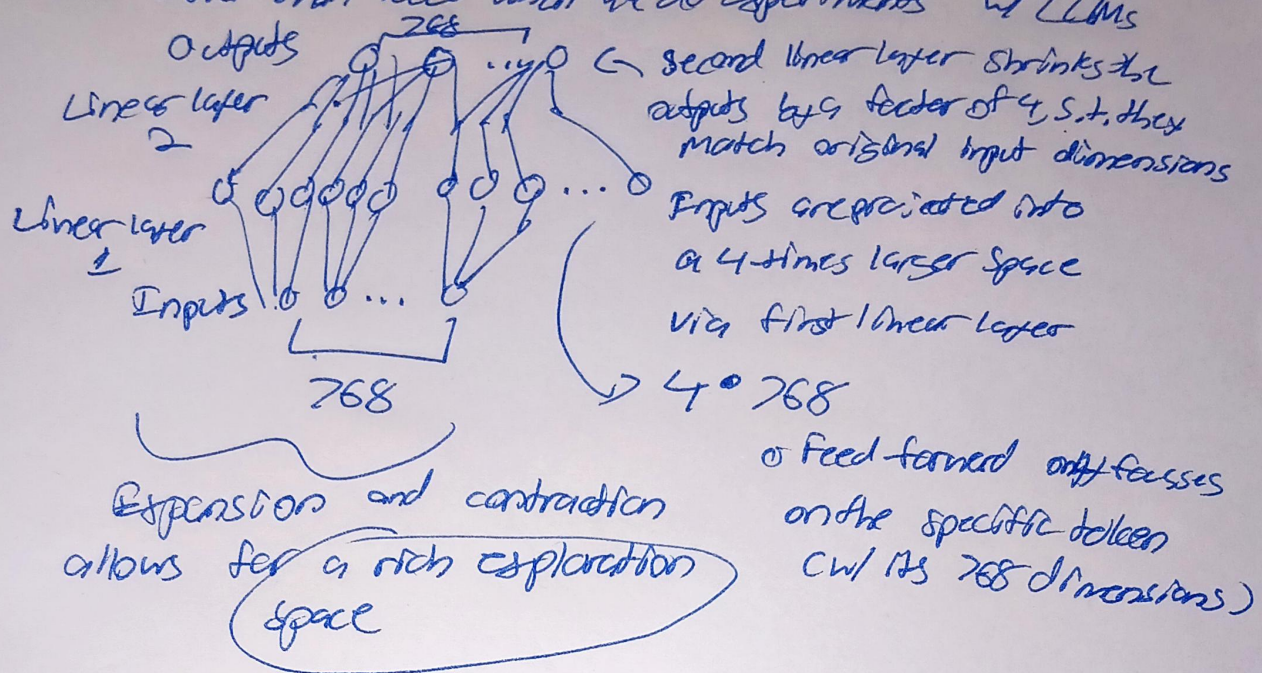
$\rightarrow$ Lets code this now

GELU Activation Function



① Using GELU activation function, lets implement the small NN model; Feed Forward that will be using in LLM transformer block

- GELU is smooth threshold, so it's differentiable for all x
- Not 0 for negative x (won't be dead; still contribute to learning)
- works better than ReLU when we do experiments w/ LLMs

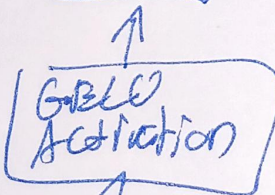


Output tensor w/
(2, 3, 768)

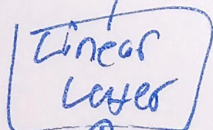


Input: (2, 3, 3072)
Output: (2, 3, 768)

Second linear layer decreases embedding dimension by a factor of 4



Input: (2, 3, 3072)
Output: (2, 3, 3072)



Input: (2, 3, 768)
Output: (2, 3, 3072)

First layer increases embedding dimension by a factor of 4

Input tensor w/
Shape (2, 3, 768)

3 values represent the batch size (2), # of tokens (3), embedding size (768)